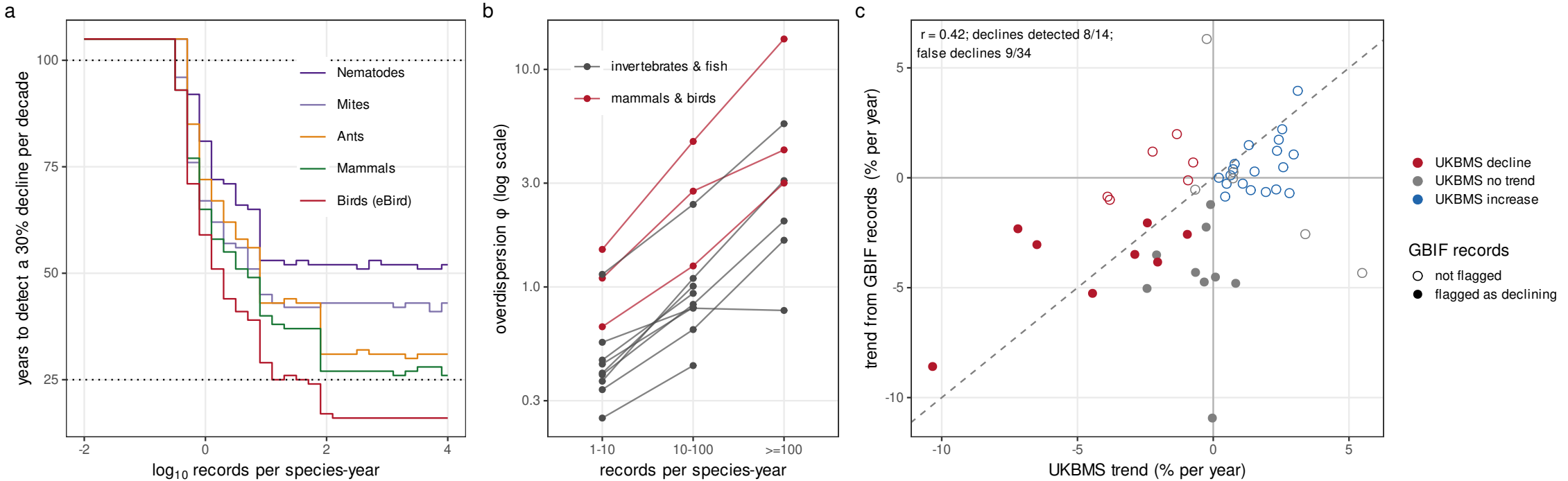

(a) Simulated years to detect a 30% decline per decade (power 0.8; negative binomial regression with effort offset; constant effort), using overdispersion estimated for each group and record rate after removing species-specific trends. 105 = not within 100 years. (b) Overdispersion of annual record counts by record rate. (c) British butterflies 2000–2019: trends from unstructured GBIF records (effort offset) vs the UK Butterfly Monitoring Scheme (48 species).